RIYADH HEATREADY

Where should planners inspect first?

Satellite and population data for southeast Riyadh

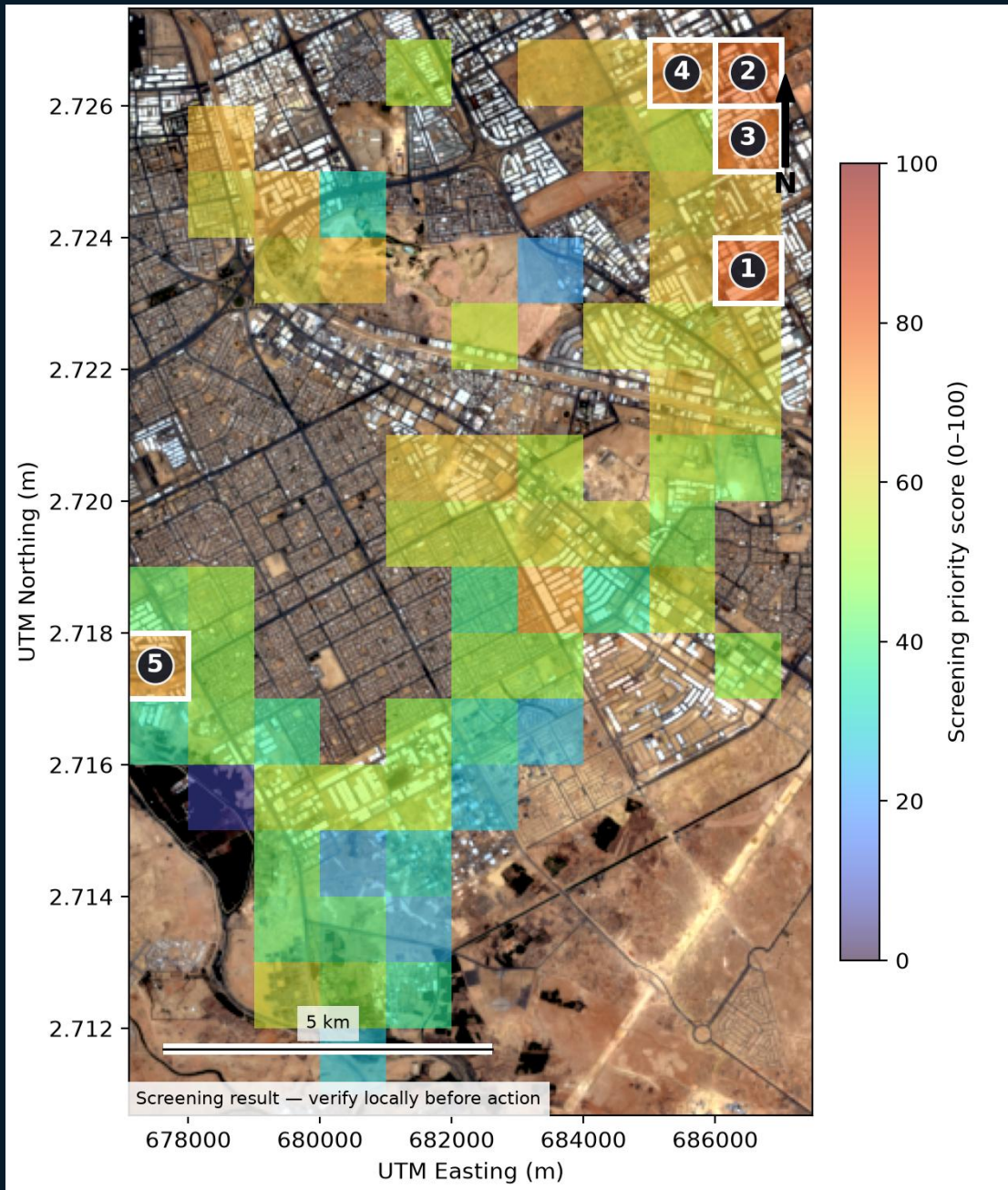
WHAT IT GIVES

A ranked map of 1 km areas that may need an early site visit

Team Milkaholics

Yusuf Karanib and Raed Mardoum

PRIORITY MAP



Our team

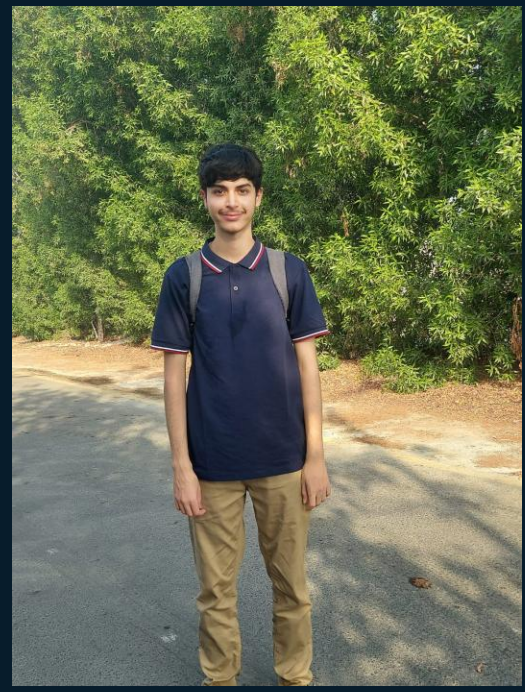


Yusuf Karanib

Early-career AI and cloud developer based in the UAE, with a background in game development and business studies in Canada.

Project lead and demo

Explains the problem, data and result. Runs the live demonstration.



Raed Mardoum

Team member focused on checking the evidence, limits and practical use of the project.

Validation and city test

Explains the validation results and how planners could test the idea.

The problem and our answer

168.3 km²	1 km	Planners	Inspect first
AREA WE STUDIED	SIZE OF EACH CELL	WHO THIS IS FOR	WHAT THE TOOL SHOWS

PROBLEM

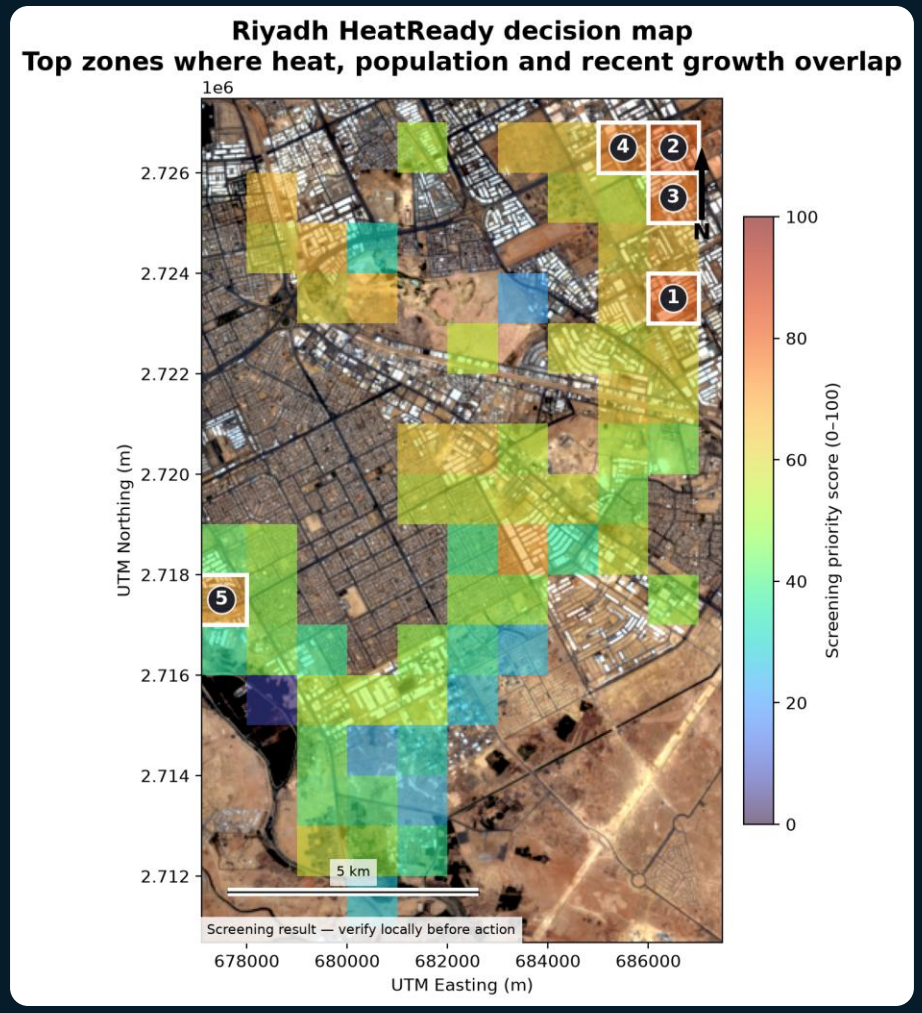
Planners may need to compare separate maps for growth, heat, population and vegetation before choosing where to visit.

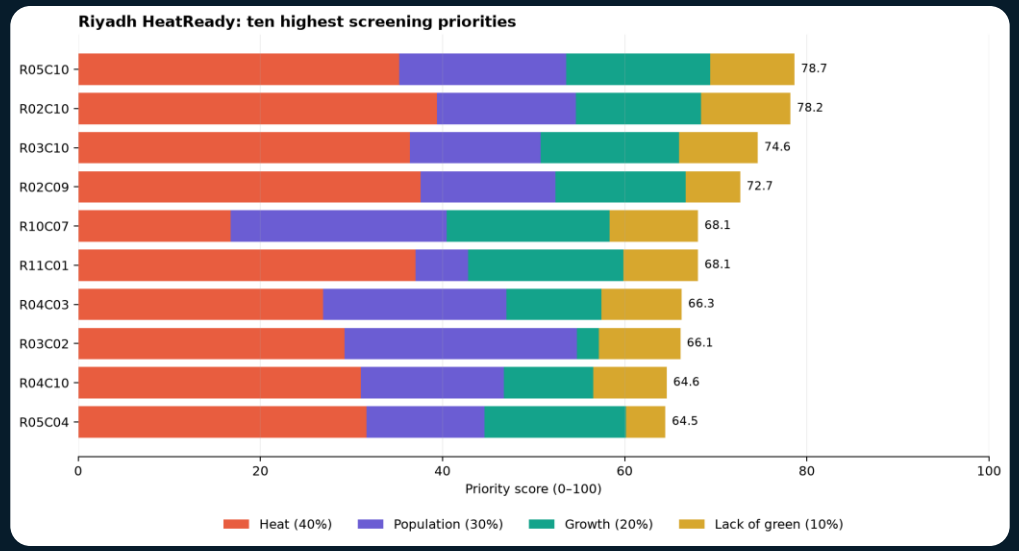
We still need to test this with real municipal users.

OUR IDEA

HeatReady combines those maps and ranks 1 km areas. It gives planners a place to start, not a final decision.

Where planners should inspect first





3.74 km²
CONFIRMED GROWTH SIGNAL

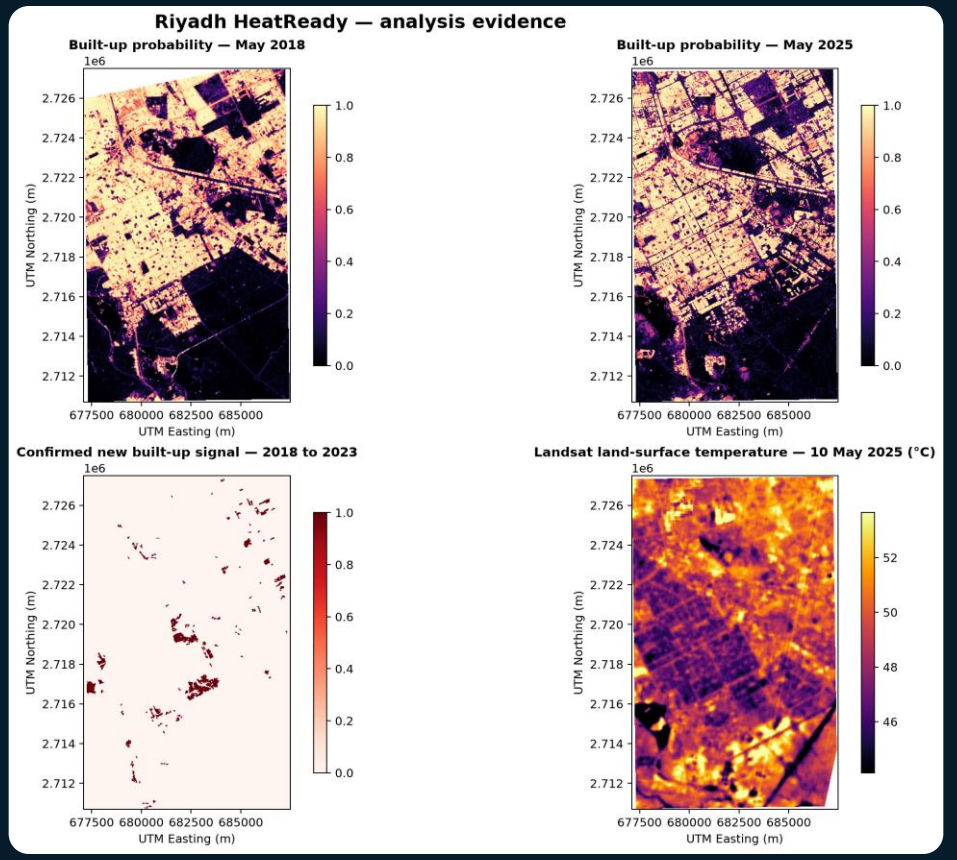
14 / 67
HIGH / ELIGIBLE CELLS

78.7
TOP SCORE: R05C10

The same result is available as a map, a table and GIS files.

The data behind the answer

Sentinel-2	Shows vegetation and surface colour
Sentinel-1	Radar check for built structures
Annual land cover	Flags possible growth from 2018 to 2023
Landsat 8	Measures land surface temperature
WorldPop	Estimates where people live
Planet Tanager	Tests a desert classification problem



Annual maps find possible growth. Optical and radar data check it. Heat, population and vegetation complete the score.

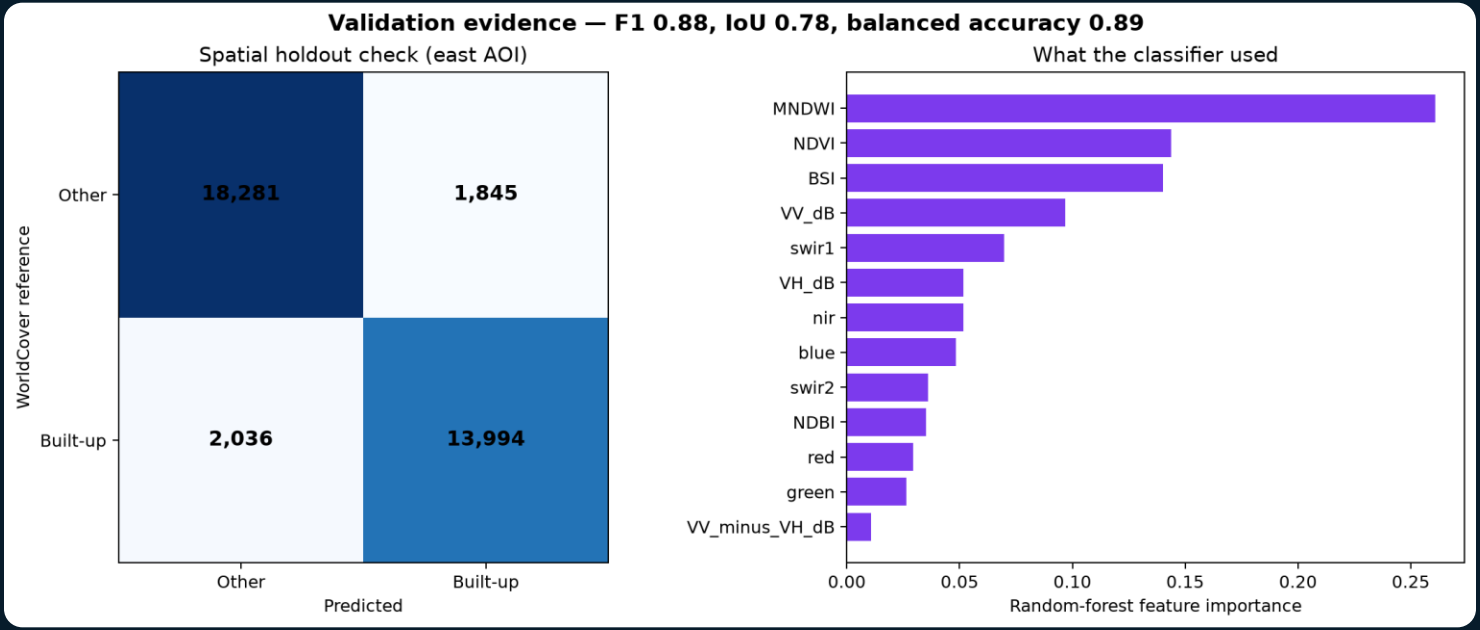
How well the model matched the reference

0.88
F1 AGREEMENT

0.78
INTERSECTION OVER UNION

0.89
BALANCED ACCURACY

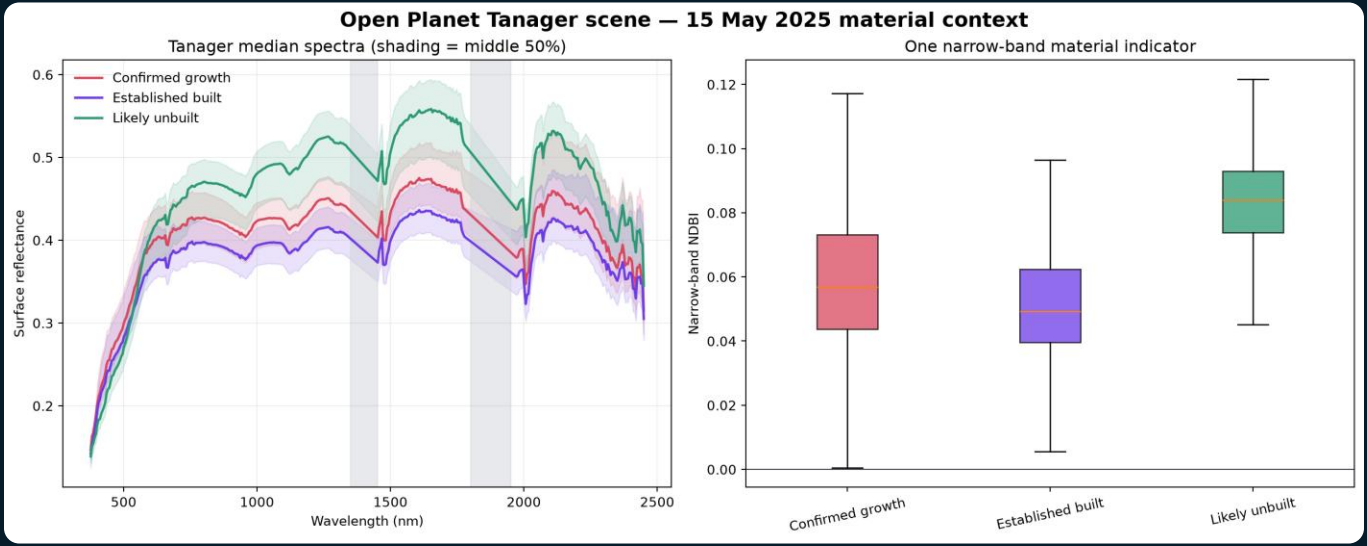
The model matched a separate satellite reference well. This is not the same as field accuracy.



MAIN LIMITATIONS

- Surface temperature is not air temperature or personal heat exposure.
- WorldPop is an estimate, not a current population or worker count.
- Bright desert and roofs can still confuse the model.
- A real user still needs to review the score weights.

Why we used more than one signal



THREE CHECKS

- Annual maps suggest where growth may have happened.
- Optical and radar data check the same places.
- Each final score keeps its four parts visible.

The Tanager test showed why one built-up index can be misleading in bright desert.

WHY IT HELPS

A planner gets one ranked list and can still see the evidence behind each area.

How we would test it with planners

1

Talk to users

Ask planners and GIS staff how they choose inspection areas today.

2

Check the top five

Compare the ranking with local plans and visits to the area.

3

Fix the score

Adjust the weights and add local information that we are missing.

4

Test again

Run the same process in another part of Riyadh.

THE TEST

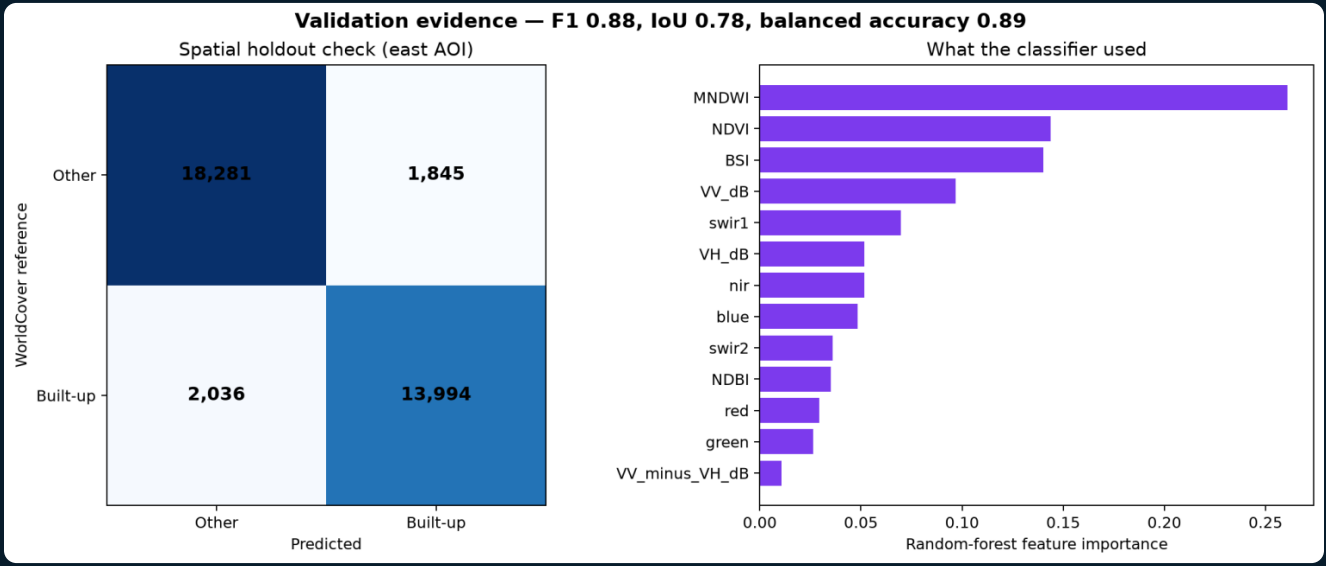
Does HeatReady save time without sending staff to the wrong places?

Data sources and dates

Source	Dates used	Role	Access / licence
Sentinel-2 L2A	11 May 2018; 10 May 2021; 9 May 2025	Optical features, masks, vegetation	Copernicus open access
Sentinel-1 RTC	10 May 2018; 6 May 2021; 9 May 2025	Radar structure	Microsoft PC; CC BY 4.0
IO/Esri annual LULC	2018-2023	Trend + candidate endpoint change	CC BY 4.0
Landsat 8 C2 L2	10 May 2025	Land-surface temperature	US public domain
ESA WorldCover	2021	Spatial holdout reference	CC BY 4.0
WorldPop Global2	2025	Modelled population	CC BY 4.0
Planet Tanager	15 May 2025	Spectral context + sensitivity test	Open urban archive; CC BY 4.0

Item IDs, processing steps and licences are documented in the repository. Satellite 813 imagery was not used.

Validation evidence and limits



Training area: west Tuning area: middle Test area: east

RESULTS

- The east-side holdout was compared with WorldCover.
- F1 0.88, IoU 0.78 and balanced accuracy 0.89.
- The test contains 36,156 reference pixels.

LIMITS

- Field accuracy.
- Air temperature or health outcomes.
- Municipal approval or user acceptance.

Next: field and user review of the top-ranked cells.